

Trios Cyber Internship — Day 01 Technical Report

Kali Linux & Network Setup

Organization: Trios Cyber

Internship: Cybersecurity Internship

Task: Kali Linux & Network Setup

Day: 01

Date: 01 September 2026

Environment: Kali Linux Virtual Machine

Virtualization Platform: Oracle VirtualBox

Network Mode: NAT

Status: Completed

1. Executive Summary

The Day 01 task focused on establishing and validating a Kali Linux virtual laboratory environment and performing basic system and network enumeration.

The assessment covered system identification, network interface enumeration, IP address identification, routing analysis, connectivity verification, and listening-service enumeration.

All required commands were executed in the Kali Linux lab environment. Command outputs were preserved as raw log files, and screenshots were captured as evidence.

2. Objectives

The objectives of this task were to:

1. Verify the Kali Linux hostname.
 2. Identify the current user.
 3. Enumerate available network interfaces.
 4. Identify the lab/attacker IP address.
 5. Identify the default gateway.
 6. Inspect the routing table.
 7. Verify local network connectivity.
 8. Verify external network connectivity.
 9. Enumerate listening TCP/UDP services.
 10. Document the results using screenshots and command logs.
-

3. Lab Environment

Parameter	Value
Operating System	Kali Linux
Environment	Virtual Machine
Virtualization	Oracle VirtualBox
Network Mode	NAT
Primary Interface	eth0
Lab/Attacker IPv4	10.0.2.15/24
Default Gateway	10.0.2.2
Loopback Address	127.0.0.1

4. System Identification

4.1 Hostname

Command:

```
hostname
```

Result:

```
kali
```

The hostname of the Kali Linux system was confirmed as `kali`.

Evidence: screenshots/01-hostname.png

4.2 Current User

Command:

```
whoami
```

Result:

```
yuvaraj
```

The current logged-in user was confirmed as `yuvaraj`.

Evidence: screenshots/02-whoami.png

5. Network Interface Enumeration

5.1 IP Address and Interface Configuration

Command:

```
ip a
```

The primary network interface identified was:

```
eth0
```

The assigned IPv4 address was:

```
10.0.2.15/24
```

The loopback interface was:

```
lo
```

```
127.0.0.1/8
```

The `eth0` interface was operational and configured with the `10.0.2.0/24` network.

Evidence: screenshots/03-ip-a.png

5.2 Legacy Interface Information

Command:

```
ifconfig
```

The `ifconfig` output confirmed:

```
Interface: eth0
```

```
IPv4 Address: 10.0.2.15
```

Netmask: 255.255.255.0
Broadcast: 10.0.2.255

The interface reported no RX or TX errors during the captured observation.

Evidence: screenshots/04-ifconfig.png

6. Routing Analysis

6.1 Routing Table

Command:

```
ip route
```

The default route was identified as:

```
default via 10.0.2.2 dev eth0
```

The local network route was:

```
10.0.2.0/24 dev eth0
```

Routing Findings

Parameter	Value
Default Gateway	10.0.2.2
Interface	eth0
Local Network	10.0.2.0/24
Source IP	10.0.2.15

The routing configuration indicates that traffic outside the local 10.0.2.0/24 network is forwarded through 10.0.2.2.

Evidence: screenshots/05-ip-route.png

7. Connectivity Verification

7.1 Gateway Connectivity

The default gateway was tested using ICMP:

```
ping -c 4 10.0.2.2
```

Result:

```
4 packets transmitted  
4 packets received  
0% packet loss
```

Average round-trip time:

```
0.445 ms
```

This confirmed successful communication between the Kali VM and its configured gateway.

Raw output: logs/ping-gateway.txt

7.2 External Network Connectivity

External connectivity was tested using Google's public DNS server:

```
ping -c 4 8.8.8.8
```

Result:

```
4 packets transmitted  
4 packets received  
0% packet loss
```

Average round-trip time:

```
7.590 ms
```

This confirmed that the Kali VM had functional external network connectivity.

Raw output: logs/ping-internet.txt

Evidence: screenshots/06-ping.png

8. Listening Service Enumeration

Listening TCP/UDP services were checked using the socket listing output.

The captured result contained only the header:

```
Netid State Recv-Q Send-Q Local Address:Port Peer Address:Port Process
```

No listening services were reported during the observation.

This indicates that no TCP/UDP listening sockets were detected at the time of enumeration.

Raw output: logs/listening-services.txt

Evidence: screenshots/07-listening-services.png

9. Network Configuration Summary

Item	Finding
Hostname	kali
Current User	yuvaraj
Primary Interface	eth0
Lab/Attacker IPv4	10.0.2.15/24
Local Network	10.0.2.0/24
Default Gateway	10.0.2.2
Loopback	127.0.0.1
Gateway Connectivity	Successful
Gateway Packet Loss	0%
External Connectivity	Successful
External Packet Loss	0%
Listening Services	None reported

10. Evidence and Documentation

The following evidence was collected during the task.

Screenshots

File	Evidence
01-hostname.png	Hostname verification
02-whoami.png	Current user verification
03-ip-a.png	IP addresses and interfaces
04-ifconfig.png	Interface configuration
05-ip-route.png	Routing table
06-ping.png	Connectivity testing
07-listening-services.png	Listening-service enumeration

Raw Logs

File	Description
day1-network-enumeration.txt	Main enumeration output
network-summary.txt	Network configuration summary
listening-services.txt	Listening-service enumeration
ping-gateway.txt	Gateway connectivity test
ping-internet.txt	External connectivity test

10. Screenshot Evidence

This section contains the visual evidence collected during the Day 01 Kali Linux and network setup exercise.

10.1 Hostname Verification

The `hostname` command was used to verify the system hostname.

10.2 Current User Verification

The `whoami` command was used to identify the active user.

10.3 Network Interface and IP Address Enumeration

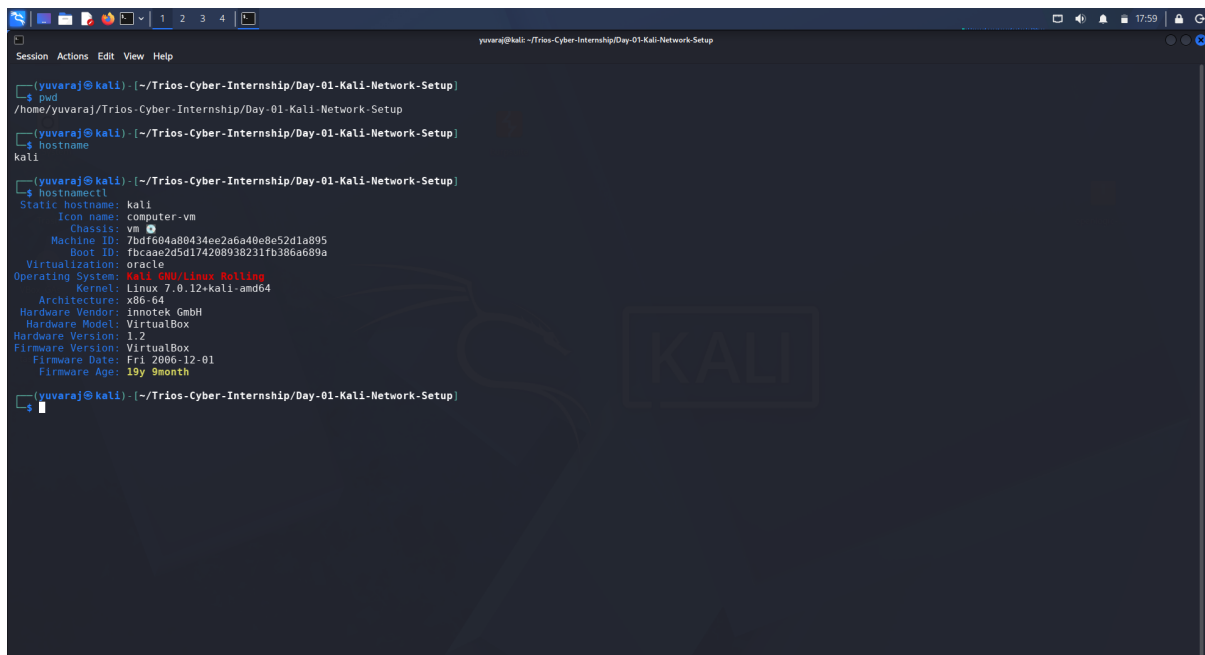
The `ip a` command was used to identify available interfaces and assigned IP addresses.

10.4 Interface Configuration

The `ifconfig` command was used to inspect interface configuration and network statistics.

10.5 Routing Table

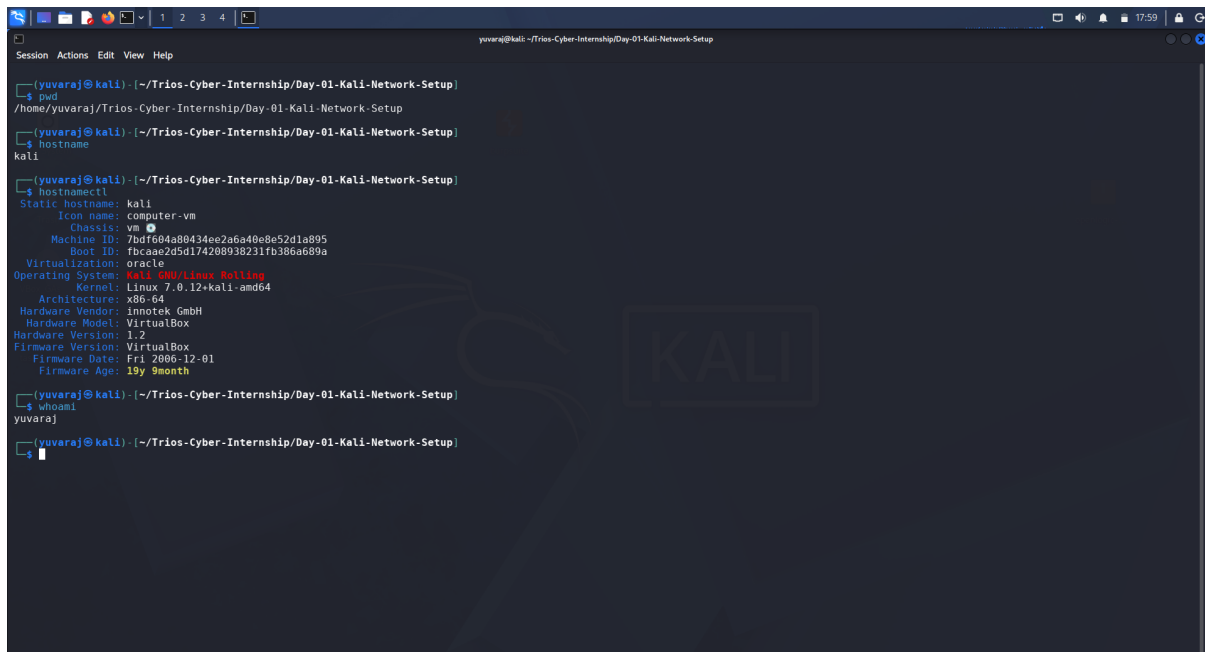
The `ip route` command was used to identify the default gateway and routing configuration.



A terminal window titled 'yuvaraj@kali: ~/Trios-Cyber-Internship/Day-01-Kali-Network-Setup'. The user runs 'hostname' and 'hostnamectl'. The output of 'hostnamectl' shows system details: Static hostname: kali, Icon name: computer-vm, Chassis: vm, Machine ID: 7bdf694a80434ee2a6a40e8e52d1a895, Boot ID: fbcaae2d5d174208938231fb386a689a, Virtualization: oracle, Operating System: Kali GNU/Linux Rolling, Kernel: Linux 7.0.12+kali-amd64, Architecture: x86_64, Hardware Vendor: innotek GmbH, Hardware Model: VirtualBox, Hardware Version: 1.2, Firmware Version: VirtualBox, Firmware Date: Fri 2006-12-01, Firmware Age: 19y 9month. A large 'KALI' watermark is visible in the background.

```
yuvaraj@kali: ~/Trios-Cyber-Internship/Day-01-Kali-Network-Setup
$ pwd
/home/yuvaraj/Trios-Cyber-Internship/Day-01-Kali-Network-Setup
$ hostname
kali
$ hostnamectl
Static hostname: kali
Icon name: computer-vm
Chassis: vm
Machine ID: 7bdf694a80434ee2a6a40e8e52d1a895
Boot ID: fbcaae2d5d174208938231fb386a689a
Virtualization: oracle
Operating System: Kali GNU/Linux Rolling
Kernel: Linux 7.0.12+kali-amd64
Architecture: x86_64
Hardware Vendor: innotek GmbH
Hardware Model: VirtualBox
Hardware Version: 1.2
Firmware Version: VirtualBox
Firmware Date: Fri 2006-12-01
Firmware Age: 19y 9month
```

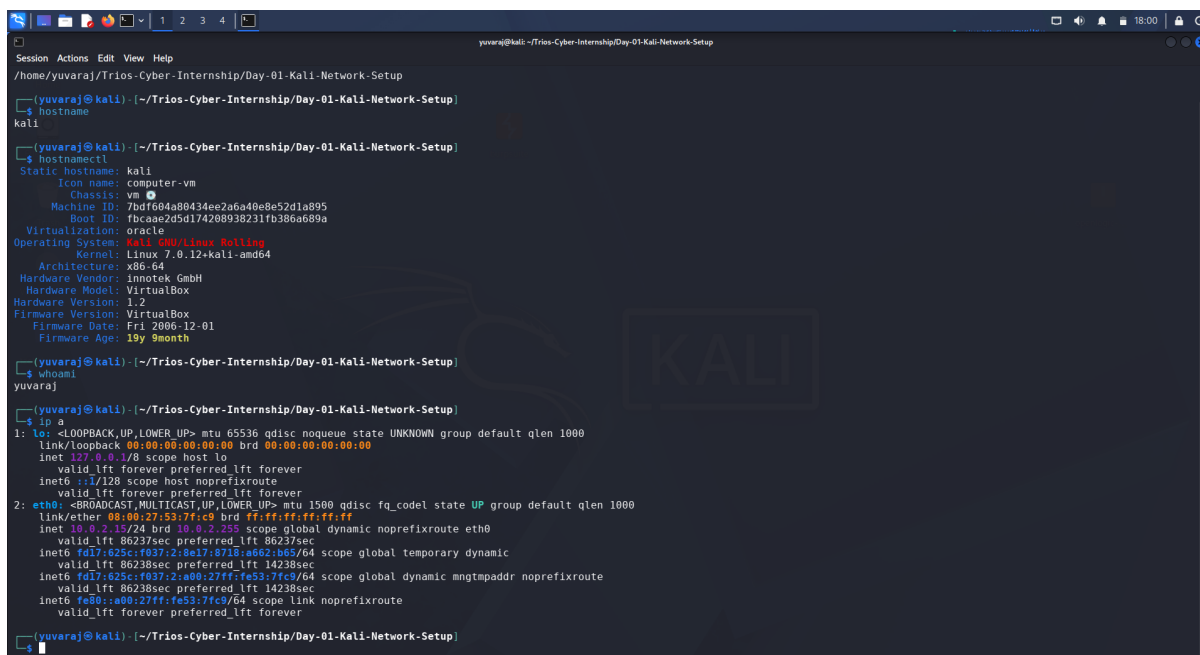
Figure 1: Hostname verification



The same terminal window as Figure 1, but now showing the output of the 'whoami' command, which returns 'yuvaraj'. The 'KALI' watermark is still present.

```
yuvaraj@kali: ~/Trios-Cyber-Internship/Day-01-Kali-Network-Setup
$ whoami
yuvaraj
```

Figure 2: Current user verification



```
Session Actions Edit View Help
/home/yuvaraj/Trios-Cyber-Internship/Day-01-Kali-Network-Setup

(yuvaraj@kali) ~/Trios-Cyber-Internship/Day-01-Kali-Network-Setup
$ hostname
kali

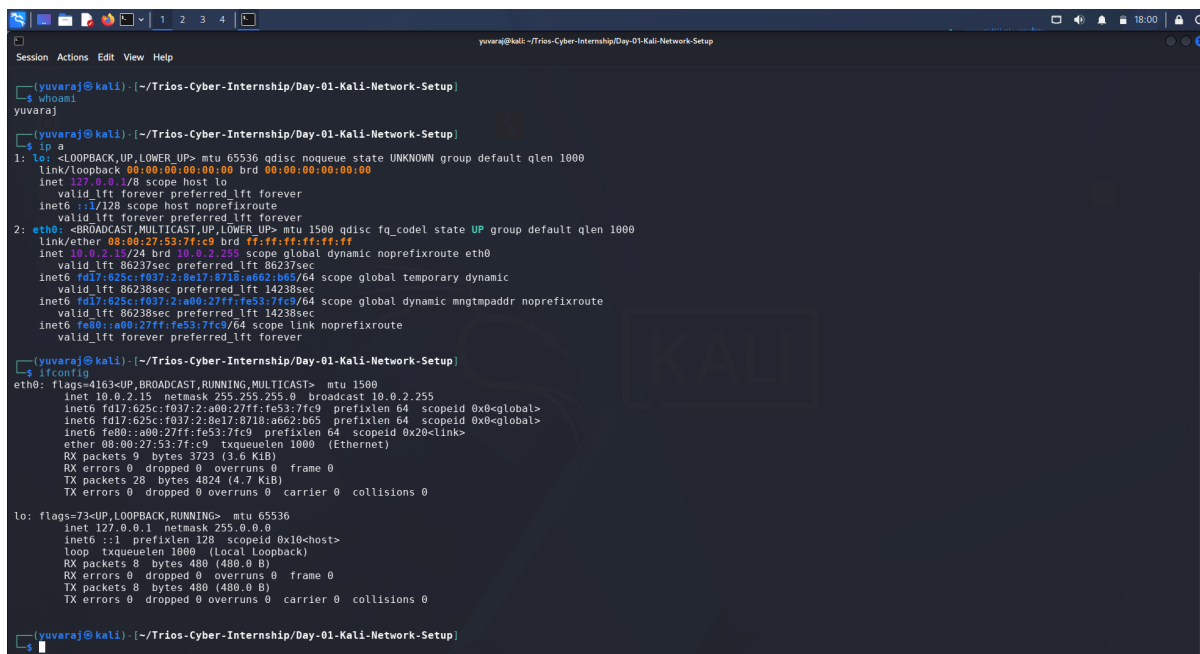
(yuvaraj@kali) ~/Trios-Cyber-Internship/Day-01-Kali-Network-Setup
$ hostnamectl
Static hostname: kali
Icon name: computer-vm
Chassis: vm
Machine ID: 7bdf604a80434ee2a6a40e8e52d1a895
Boot ID: fbcaae2d5d174208938231fb386a689a
Virtualization: oracle
Operating System: Kali GNU/Linux Rolling
Kernel: Linux 7.0.12-kali-amd64
Architecture: x86_64
Hardware Vendor: innovek GmbH
Hardware Model: VirtualBox
Hardware Version: 1.2
Firmware Version: VirtualBox
Firmware Date: Fri 2006-12-01
Firmware Age: 19y 9month

(yuvaraj@kali) ~/Trios-Cyber-Internship/Day-01-Kali-Network-Setup
$ whoami
yuvaraj

(yuvaraj@kali) ~/Trios-Cyber-Internship/Day-01-Kali-Network-Setup
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:53:7f:c9 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
        valid_lft 86237sec preferred_lft 86237sec
    inet6 fd17:625c:f037:2:a00:27ff:fe53:7fc9/64 scope global temporary dynamic
        valid_lft 86238sec preferred_lft 14238sec
    inet6 fd17:625c:f037:2:a00:27ff:fe53:7fc9/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 86238sec preferred_lft 14238sec
    inet6 fe80::a00:27ff:fe53:7fc9/64 scope link noprefixroute
        valid_lft forever preferred_lft forever

(yuvaraj@kali) ~/Trios-Cyber-Internship/Day-01-Kali-Network-Setup
$
```

Figure 3: IP address and interface enumeration



```
Session Actions Edit View Help
/home/yuvaraj/Trios-Cyber-Internship/Day-01-Kali-Network-Setup

(yuvaraj@kali) ~/Trios-Cyber-Internship/Day-01-Kali-Network-Setup
$ whoami
yuvaraj

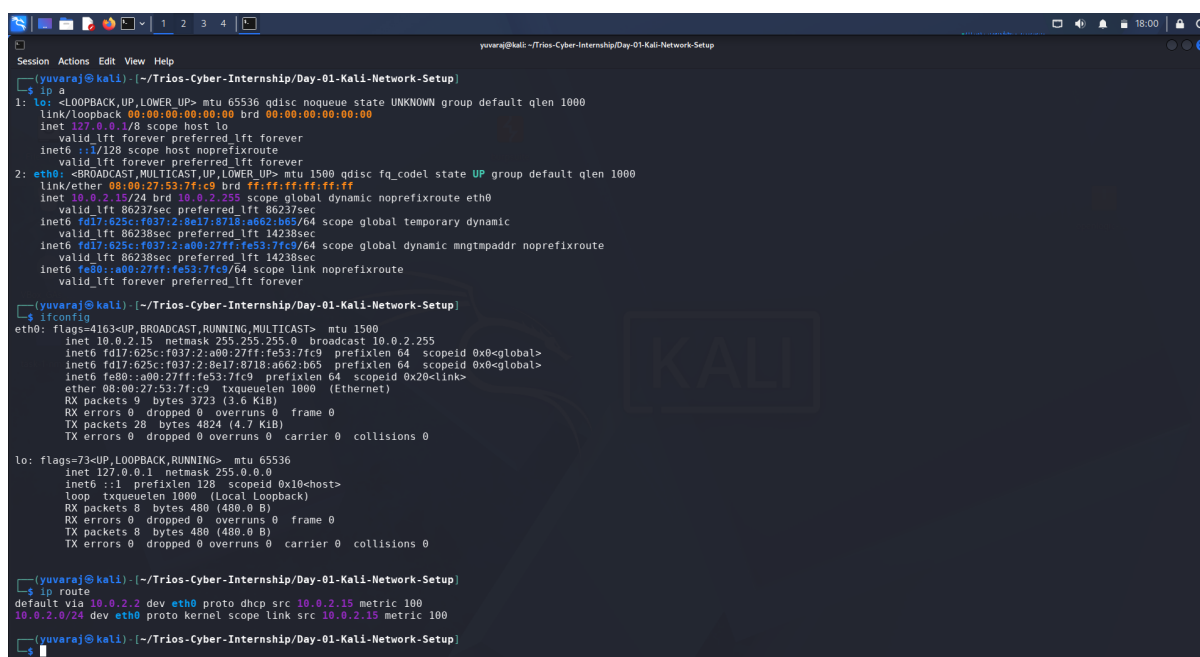
(yuvaraj@kali) ~/Trios-Cyber-Internship/Day-01-Kali-Network-Setup
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:53:7f:c9 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
        valid_lft 86237sec preferred_lft 86237sec
    inet6 fd17:625c:f037:2:a00:27ff:fe53:7fc9/64 scope global temporary dynamic
        valid_lft 86238sec preferred_lft 14238sec
    inet6 fd17:625c:f037:2:a00:27ff:fe53:7fc9/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 86238sec preferred_lft 14238sec
    inet6 fe80::a00:27ff:fe53:7fc9/64 scope link noprefixroute
        valid_lft forever preferred_lft forever

(yuvaraj@kali) ~/Trios-Cyber-Internship/Day-01-Kali-Network-Setup
$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.2.15 netmask 255.255.0 broadcast 10.0.2.255
    inet6 fd17:625c:f037:2:a00:27ff:fe53:7fc9 prefixlen 64 scopeid 0x0<global>
    inet6 fd17:625c:f037:2:a00:27ff:fe53:7fc9 prefixlen 64 scopeid 0x0<global>
    inet6 fe80::a00:27ff:fe53:7fc9 prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:53:7f:c9 txqueuelen 1000 (Ethernet)
    RX packets 9 bytes 3723 (3.6 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 28 bytes 4824 (4.7 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 8 bytes 480 (480.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 8 bytes 480 (480.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

(yuvaraj@kali) ~/Trios-Cyber-Internship/Day-01-Kali-Network-Setup
$
```

Figure 4: Interface configuration



```

(yuvaraj@kali) (~/.Trios-Cyber-Internship/Day-01-Kali-Network-Setup)
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:53:7f:c9 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
        valid_lft 86237sec preferred_lft 86237sec
    inet6 fd17:625c:f037:2:a00:27ff:fe53:7fc9/64 scope global temporary dynamic
        valid_lft 86238sec preferred_lft 14238sec
    inet6 fd17:625c:f037:2:a00:27ff:fe53:7fc9/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 86238sec preferred_lft 14238sec
    inet6 fe80::a00:27ff:fe53:7fc9/64 scope link noprefixroute
        valid_lft forever preferred_lft forever

(yuvaraj@kali) (~/.Trios-Cyber-Internship/Day-01-Kali-Network-Setup)
$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255
    inet6 fd17:625c:f037:2:a00:27ff:fe53:7fc9 prefixlen 64 scopeid 0x8<global>
    inet6 fd17:625c:f037:2:a00:27ff:fe53:7fc9 prefixlen 64 scopeid 0x8<global>
    inet6 fe80::a00:27ff:fe53:7fc9 prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:53:7f:c9 txqueuelen 1000 (Ethernet)
    RX packets 9 bytes 3723 (3.6 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 28 bytes 4824 (4.7 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

(yuvaraj@kali) (~/.Trios-Cyber-Internship/Day-01-Kali-Network-Setup)
$ ip route
default via 10.0.2.2 dev eth0 proto dhcp src 10.0.2.15 metric 100
10.0.2.0/24 dev eth0 proto kernel scope link src 10.0.2.15 metric 100

(yuvaraj@kali) (~/.Trios-Cyber-Internship/Day-01-Kali-Network-Setup)

```

Figure 5: Routing table

10.6 Connectivity Verification

Connectivity to the local gateway and external network was verified using ICMP ping tests.

10.7 Listening Service Enumeration

Listening TCP/UDP sockets were checked to identify services exposed by the Kali lab system.

11. Conclusion

The Day 01 Kali Linux and network setup task was successfully completed.

The lab environment was validated by confirming the hostname, active user, network interfaces, IPv4 configuration, default gateway, routing table, network connectivity, and listening services.

The primary lab interface was identified as `eth0`, with IPv4 address `10.0.2.15/24` and default gateway `10.0.2.2`.

Connectivity to both the local gateway and external IP address `8.8.8.8` was successfully verified with 0% packet loss.

No TCP/UDP listening services were detected during the service enumeration performed at the time of testing.

All command outputs and screenshots have been preserved as supporting evidence within the project repository.

11. Commands Practiced

The following Linux networking and system-identification commands were used:

```
Session Actions Edit View Help
inet6 fd17:625c:f037:2:8e17:8718:a662:b65/64 scope global temporary dynamic
valid_lft 86238sec preferred_lft 14238sec
inet6 fd17:625c:f037:2:a00:27ff:fe53:7fc9/64 scope global dynamic mngtaddr noprefixroute
valid_lft 86238sec preferred_lft 14238sec
inet6 fe80::a00:27ff:fe53:7fc9/64 scope link noprefixroute
valid_lft forever preferred_lft forever

[yuvaraj@kali] ~/Trios-Cyber-Internship/Day-01-Kali-Network-Setup
$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255
    inet6 fd17:625c:f037:2:a00:27ff:fe53:7fc9 prefixlen 64 scopeid 0x0<global>
    inet6 fd17:625c:f037:2:8e17:8718:a662:b65 prefixlen 64 scopeid 0x0<global>
    inet6 fe80::a00:27ff:fe53:7fc9 prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:53:7f:c9 txqueuelen 1000 (Ethernet)
    RX packets 9 bytes 3723 (3.6 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 20 bytes 4824 (4.7 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 8 bytes 480 (480.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 8 bytes 480 (480.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

[yuvaraj@kali] ~/Trios-Cyber-Internship/Day-01-Kali-Network-Setup
$ ip route
default via 10.0.2.2 dev eth0 proto dhcp src 10.0.2.15 metric 100
10.0.2.0/24 dev eth0 proto kernel scope link src 10.0.2.15 metric 100

[yuvaraj@kali] ~/Trios-Cyber-Internship/Day-01-Kali-Network-Setup
$ ping -c 4 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data:
64 bytes from 8.8.8.8: icmp_seq=1 ttl=64 time=0.70 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=64 time=0.43 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=64 time=0.18 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=64 time=0.24 ms

... 8.8.8.8 ping statistics ...
4 packets transmitted, 4 received, 0% packet loss, time 3105ms
rtt min/avg/max/mdev = 0.103/0.637/9.697/0.618 ms

[yuvaraj@kali] ~/Trios-Cyber-Internship/Day-01-Kali-Network-Setup
$
```

Figure 6: Connectivity verification

```
Session Actions Edit View Help
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255
    inet6 fd17:625c:f037:2:a00:27ff:fe53:7fc9 prefixlen 64 scopeid 0x0<global>
    inet6 fd17:625c:f037:2:8e17:8718:a662:b65 prefixlen 64 scopeid 0x0<global>
    inet6 fe80::a00:27ff:fe53:7fc9 prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:53:7f:c9 txqueuelen 1000 (Ethernet)
    RX packets 9 bytes 3723 (3.6 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 20 bytes 4824 (4.7 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 8 bytes 480 (480.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 8 bytes 480 (480.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

[yuvaraj@kali] ~/Trios-Cyber-Internship/Day-01-Kali-Network-Setup
$ ip route
default via 10.0.2.2 dev eth0 proto dhcp src 10.0.2.15 metric 100
10.0.2.0/24 dev eth0 proto kernel scope link src 10.0.2.15 metric 100

[yuvaraj@kali] ~/Trios-Cyber-Internship/Day-01-Kali-Network-Setup
$ ping -c 4 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data:
64 bytes from 8.8.8.8: icmp_seq=1 ttl=64 time=0.70 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=64 time=0.43 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=64 time=0.18 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=64 time=0.24 ms

... 8.8.8.8 ping statistics ...
4 packets transmitted, 4 received, 0% packet loss, time 3105ms
rtt min/avg/max/mdev = 0.103/0.637/9.697/0.618 ms

[yuvaraj@kali] ~/Trios-Cyber-Internship/Day-01-Kali-Network-Setup
$ sudo ss -tlnlp
[sudo] password for yuvaraj:
Netid      State      Recv-Q     Send-Q     Local Address:Port      Peer Address:Port      Process
[yuvaraj@kali] ~/Trios-Cyber-Internship/Day-01-Kali-Network-Setup
$ sudo ss -tulnp | tee logs/listening-services.txt
Netid State Recv-Q Send-Q Local Address:Port Peer Address:Port Process
[yuvaraj@kali] ~/Trios-Cyber-Internship/Day-01-Kali-Network-Setup
$
```

Figure 7: Listening service enumeration

```
hostname
whoami
ip a
ifconfig
ip route
ping -c 4 10.0.2.2
ping -c 4 8.8.8.8
```

Listening services were also enumerated using the socket listing functionality.

These commands provided practical exposure to system identification, interface discovery, routing analysis, connectivity testing, and basic service enumeration.

12. Conclusion

The Day 01 Kali Linux and Network Setup task was completed successfully in a dedicated VirtualBox-based Kali Linux laboratory environment.

The Kali VM was identified, its network interfaces and IP configuration were enumerated, the default gateway and routing table were identified, and network connectivity was successfully verified.

The environment demonstrated:

- A functional `eth0` interface.
- Lab IP address `10.0.2.15/24`.
- Default gateway `10.0.2.2`.
- Successful gateway connectivity with 0% packet loss.
- Successful external connectivity with 0% packet loss.
- No listening TCP/UDP services reported during the observation.

All required evidence was organized into screenshots and raw command-output logs for reproducibility and review.

13. Repository Structure

```
Day-01-Kali-Network-Setup/
  README.md
  DAY-01-REPORT.md
  logs/
    day1-network-enumeration.txt
    listening-services.txt
    network-summary.txt
    ping-gateway.txt
    ping-internet.txt
  screenshots/
    01-hostname.png
    02-whoami.png
    03-ip-a.png
    04-ifconfig.png
    05-ip-route.png
    06-ping.png
    07-listening-services.png
```

Prepared for: Trios Cyber Internship

Task: Day 01 — Kali Linux & Network Setup